Omkar Chittar

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Work Experience

NBCUniversal Orlando, FL

Lead Software Engineer *Nov 2025 – Present*

- Independently designed and built a **full-stack estimating and takeoff platform** (Python backend, **React/TypeScript** frontend): document-ingestion pipeline, OCR, spatial registration, and a geometry-aware computation engine with AI-assisted extraction; cut takeoff time from **6 hours to 30 minutes**, backed by **∼900 automated tests**
- Led software delivery across **Universal Creative** and **UDX**, partnering with stakeholders across **6 business functions** to gather requirements and author PRDs, shipping **10 products** from concept into production
- Architected **8+ production systems** spanning estimating, legal, engineering, safety, finance, and operations, with shared patterns for service routing, validation, access control, and **human-in-the-loop approval**
- Led development of a centralized **contract lifecycle platform**, building ETL and a unified **search layer over 10,000+ documents** from SharePoint, databases, and enterprise repositories; reduced first-draft contract generation from ∼2 weeks to **under 48 hours**
- Built retrieval and evaluation infrastructure over 5,000+ pages of legal documents with semantic search, automated citation validation, and deterministic checks; achieved full citation coverage at a **∼2% review-error rate**
- Delivered a **cost-intelligence** platform over historical cost data with structured retrieval, query routing, provenance tracking, and a conversational UI; reduced cost-research time **95%**
- Built and deployed a production **CI debugging agent** using Python, FastAPI, GitHub Actions, and LLM tool calling to analyze failed builds, identify likely root causes, and recommend code fixes

FOX Sports Redmond, WA (Remote)

AI Engineer *Aug 2024 – Nov 2025*

- Built and deployed **production video understanding pipelines** on GCP using Python, FFmpeg, CLIP/SigLIP, and TimeSformer, generating **100K+ searchable clip embeddings monthly**; integrated FAISS/HNSW and BigQuery for semantic retrieval, contributing to **30% higher retrieval accuracy** across the multimodal search platform.
- Engineered **production speech transcription and speaker diarization pipelines** using Python, FFmpeg, Whisper, VAD, and speaker-embedding clustering, processing **800+ hours of audio monthly**; generated timestamp-aligned, speaker-labeled transcripts and multilingual metadata, contributing to a **40% reduction in manual media search effort** across the platform.
- Decreased **issue resolution time 35%** by building an event-driven **monitoring service** (AWS Bedrock, Lambda, Slack API) for ML pipelines, surfacing drift alerts, exceptions, and health insights in real time
- Reduced **ad-hoc reporting effort 40%** by building a **natural-language-to-SQL** assistant (FAISS, Bedrock) serving producers at **p95 ∼1.3s** latency
- Built an automated **evaluation and regression harness** using synthetic queries, oracle sets, and precision@k in the iteration workflow; improved top-5 retrieval by **+12 p.p.**
- Cut per-request serving cost **28%** without quality loss via response caching and lightweight request routing on Databricks
- Engineered and deployed **production Apache Airflow DAGs** on GCP for **100K+ video segments monthly**: FFmpeg preprocessing, Whisper transcription, CLIP/SigLIP embeddings, and BigQuery ETL; parallel execution, automated retries, SLA monitoring, and idempotent backfills achieved **99%+ pipeline success** and **50% fewer manual interventions**
- Streamlined operations on **2M+ digital assets** by automating metadata extraction and **ETL** (BigQuery, Spark, Cloud Functions); features powered engagement forecasting (LSTM, transformers) that lifted **user retention 25%**

Sakar Robotics Pune, India
Software Engineer *June 2019 – June 2022*

- Developed reusable **computer-vision** and **3D-processing modules** using OpenCV, Open3D, stereo depth, ICP, and multi-camera calibration; improved robotic pick success **18%** and reduced pose error **22%**
- Implemented TSDF mapping & 6-DoF tracking services, lowering re-localization time during occlusions by **35%**
- Developed **reinforcement-learning** components for robotic manipulation using **DDPG + HER**, improving task precision by **30%**

Skills

Languages: Python, JavaScript/TypeScript, SQL, C++
Backend & Web: Node.js, Flask, FastAPI, REST APIs, gRPC, React, Next.js
Cloud & Data: AWS (Bedrock, Lambda, SageMaker, S3), GCP (BigQuery, Vertex AI, Cloud Functions), Azure, Databricks, Snowflake, Spark, Apache Airflow, Kafka, Postgres, NoSQL, Redis, SQL query tuning, warehouse data modeling
MLOps/DevOps: MLflow, Docker, Kubernetes, CI/CD, Git/GitHub, GitHub Actions, Linux, unit/integration/end-to-end testing, vector stores & databases
GenAI/LLMs: RAG, agentic AI & tool calling, vector search, embeddings, guardrails, LLM evaluation (Ragas), prompt engineering & caching, LangChain, LangGraph, OpenAI, Claude, Hugging Face, MCP
ML/DL: NLP, Computer Vision, Whisper, CLIP/SigLIP, TimeSformer, PyTorch, TensorFlow

Publications

Two peer-reviewed papers on exoskeleton systems in *Materials Today: Proceedings*: paper 1, paper 2

Education

University of Maryland College Park, MD
Master of Engineering in Robotics, 3.97 CGPA *Aug 2022 – May 2024*